

SEIZURE SENSE

Problem

Approximately 60 million people around the world suffer from seizures and it can happen to anyone of all ages from infants to elderly. The problem is that seizures are unpredictable and they often can happen without any warning. This can be extremely dangerous as the patient may fall and injure themselves. The individual can lose consciousness, muscle contractions that can hit objects, and also falling over.

Purpose

My older sister has suffered from epilepsy for the past 10 years. This device will alert the individual as well as people around them to get them to safety. Additionally, this also gives the individual the ability to participate in activities individually. It provides a sense of protectiveness to one's family as well since they will be acknowledged when a seizure is about to occur. Anyone can witness a seizure and may not know how to help as well, these devices showcase warnings of what to do in case of an emergency.

Methodology

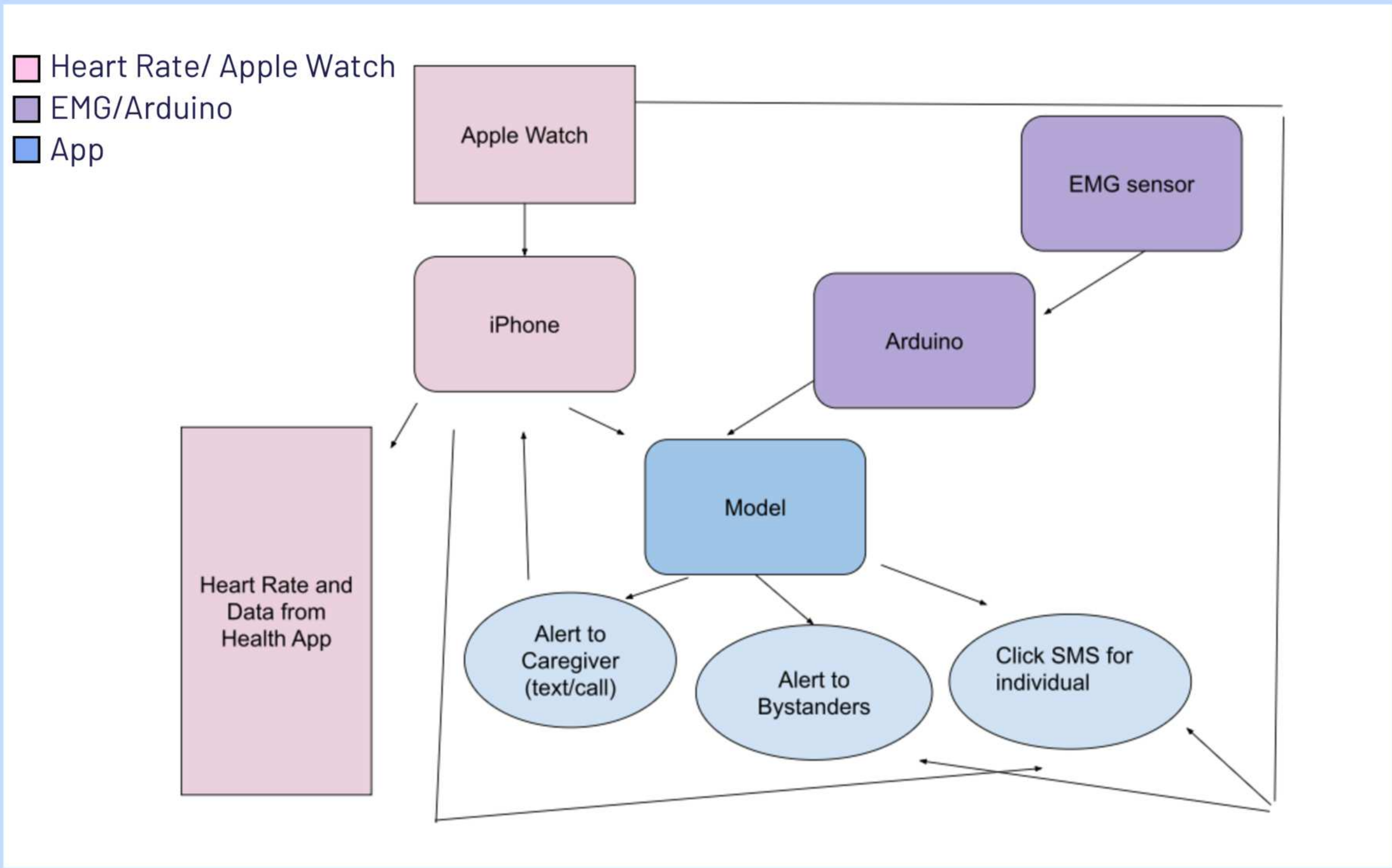


Photo created by the researcher using Google Draw

Results

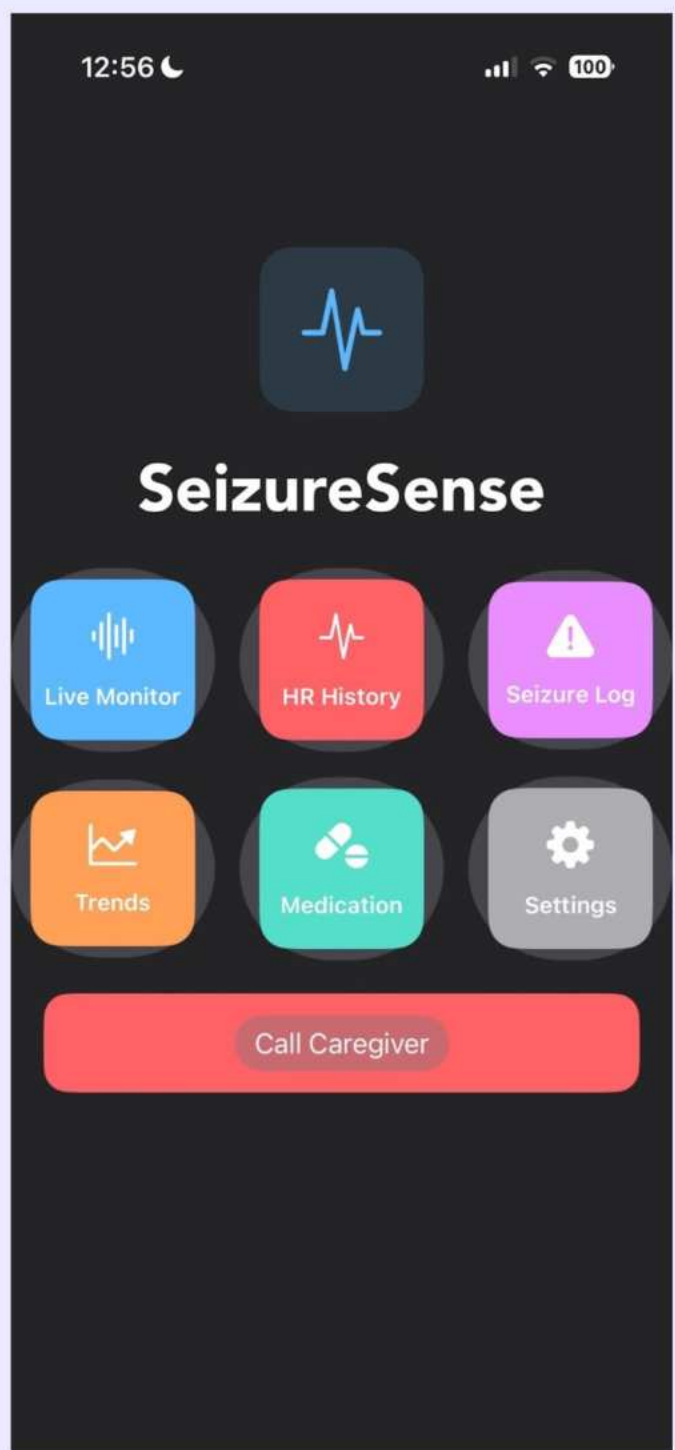


Photo created by the researcher by Xcode

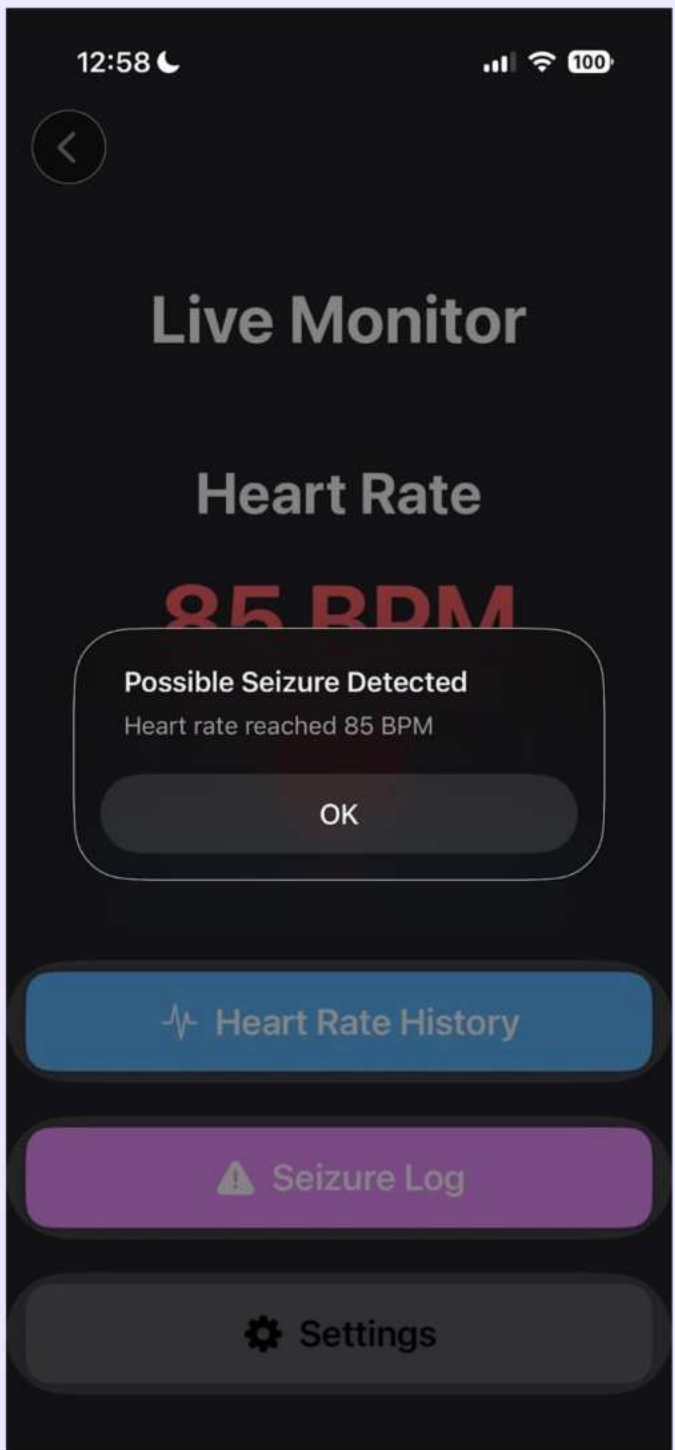


Photo created by the researcher by Xcode

The Home Screen consists of the Live monitor, HR history, Seizure Log, Trends, Medication, Settings, and Emergency Call feature. In the live monitor, the heart rate is measured and a notification pops up when a seizure is detected due to excessive heart rate

The app logs seizure data and lists the symptoms as to why seizure was detected. The heart rate history is logged with the BPM and the date therefore the user can check for abnormalities.

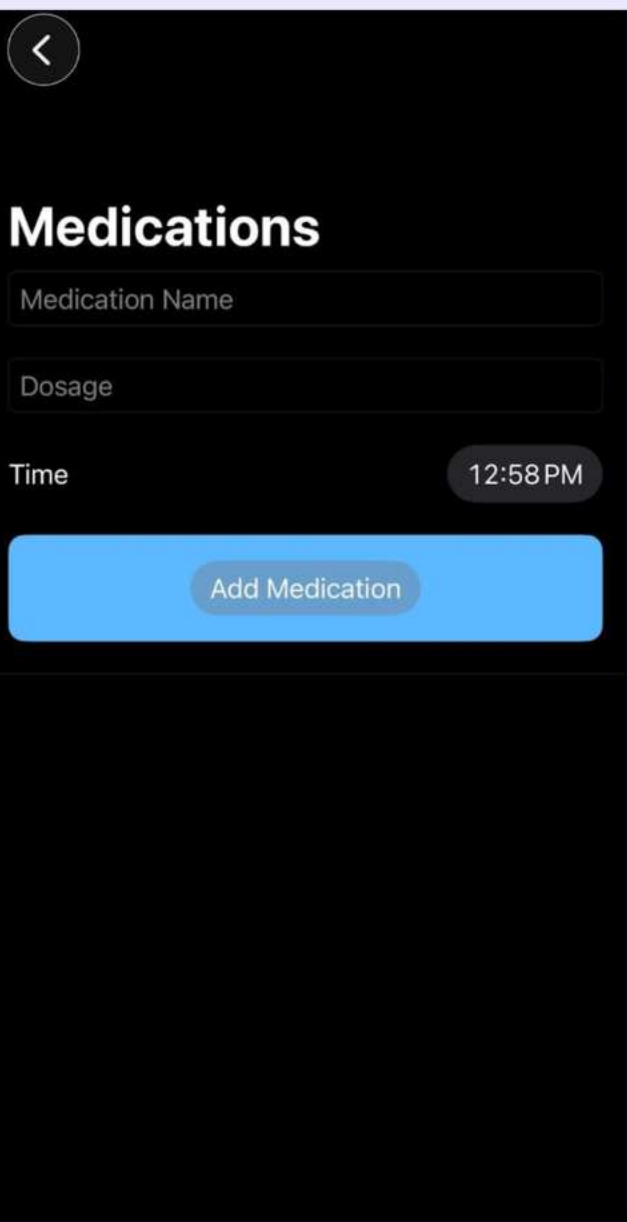


Photo created by the researcher by Xcode

The Medications section is for the user to input prescribed medication information such as the name, dosage, and time.

Using an EMG sensor to capture muscle contractions, surface electrodes are placed on the skin, specifically the upper arm. If a number of contractions is repeated at a specific threshold it will be triggered.

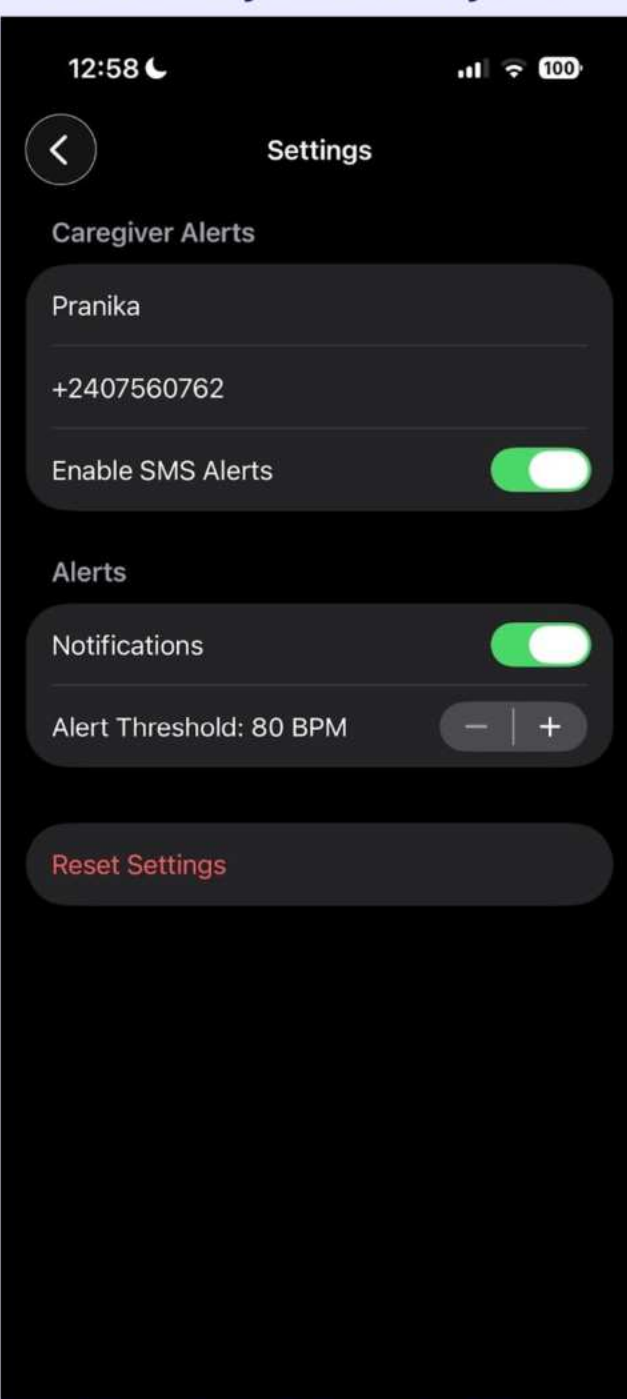


Photo created by the researcher by Xcode

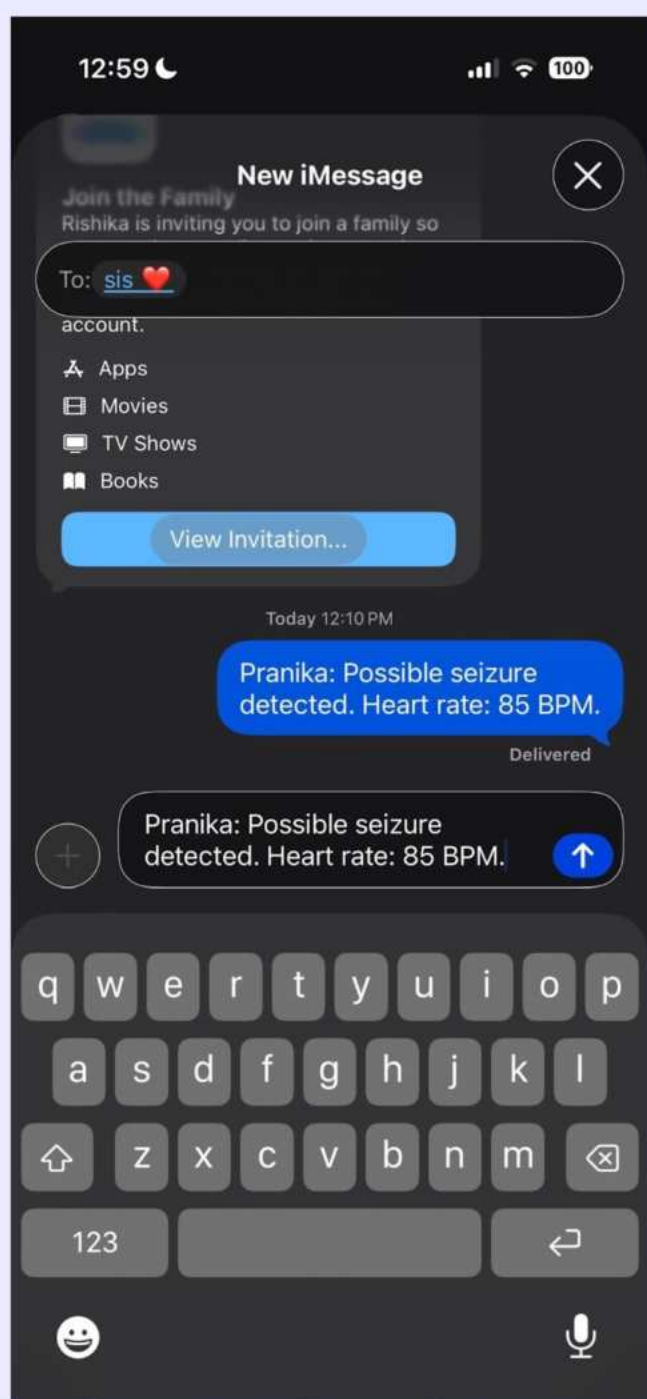


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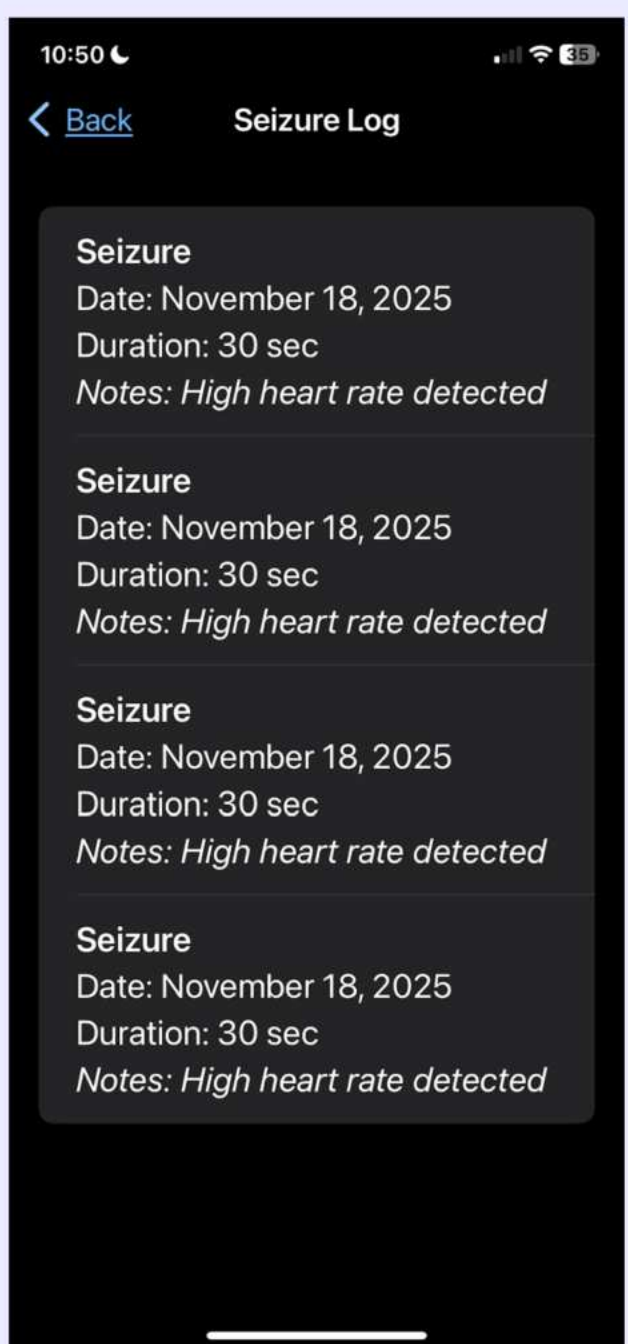


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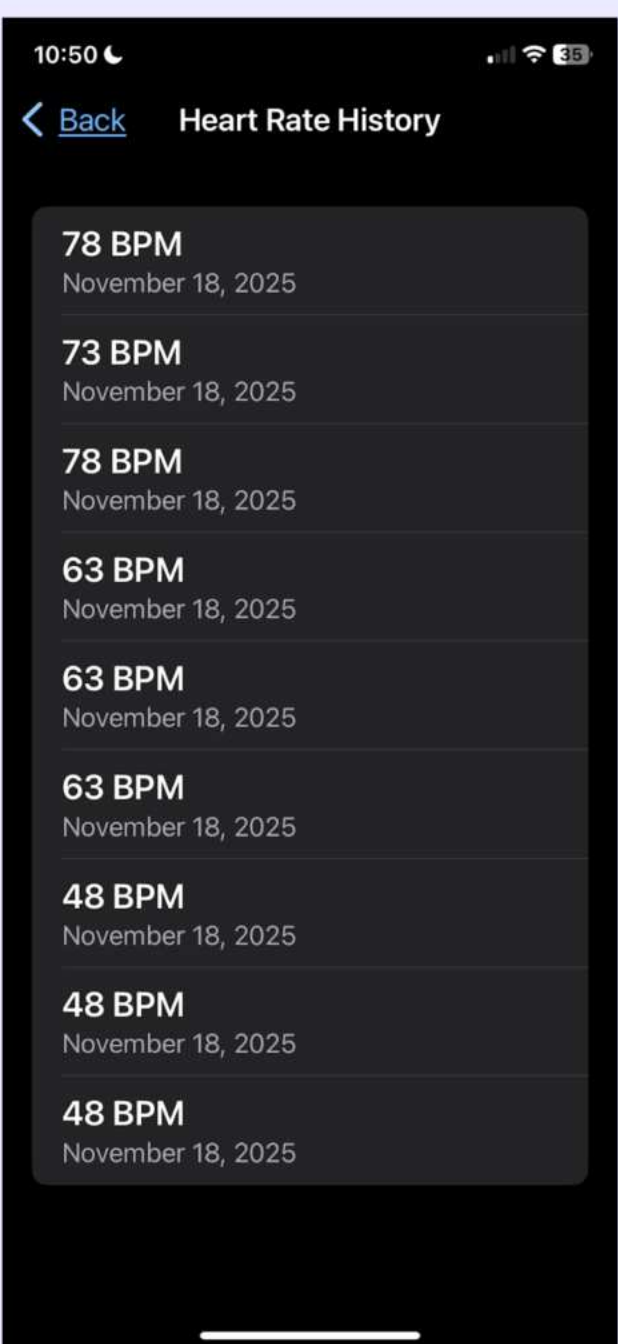


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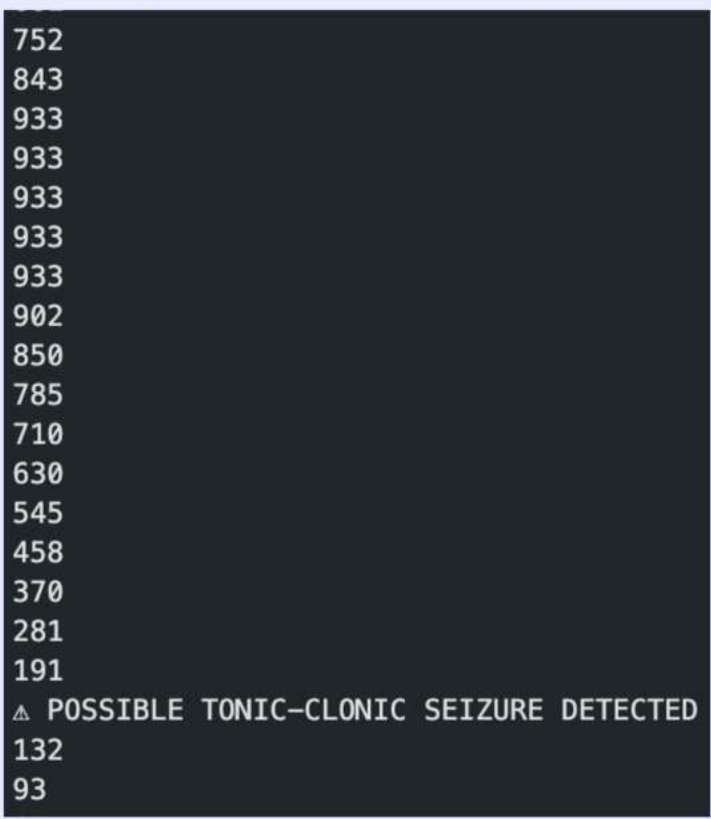


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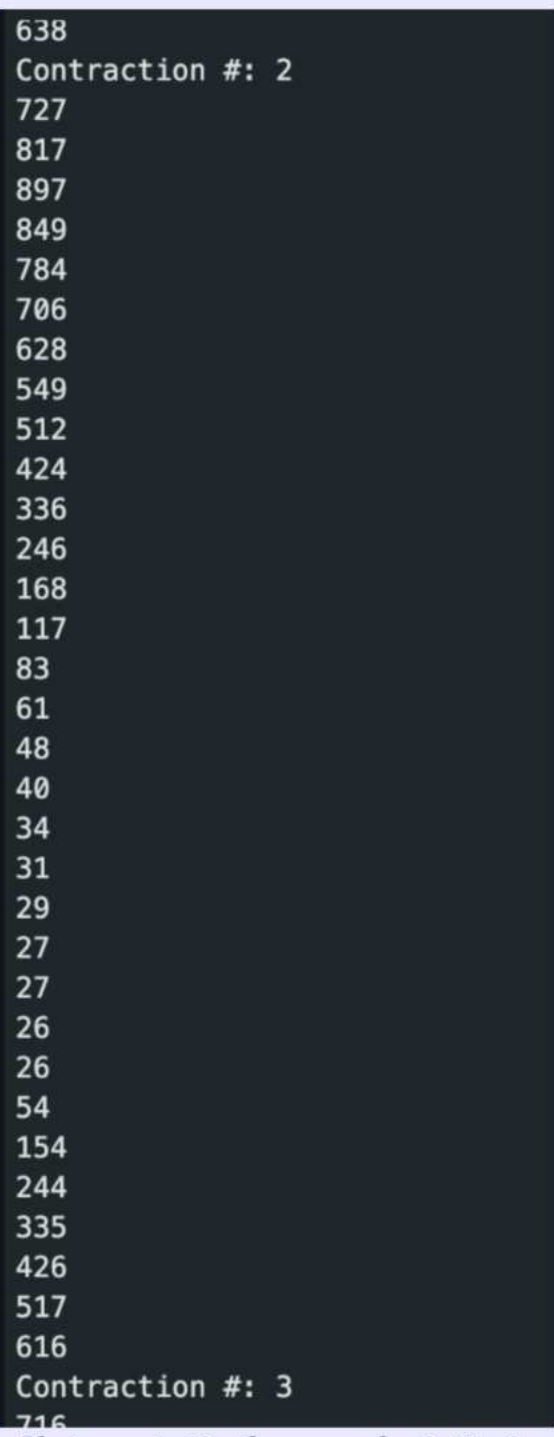


Photo created by the researcher by Xcode



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The heart rate can be modified in settings to a certain threshold after assessing heart rate changes through the graph. It is set to a default of 140 BPM which is the average heart rate during tonic-clonic seizures. Additionally, there is a caregiver alerts section where you can add a parent/guardian whom you want to receive texts or calls if a possible seizure is detected.

Results



Photo created by the researcher by Arduino

Spiked EMG

Normal EMG

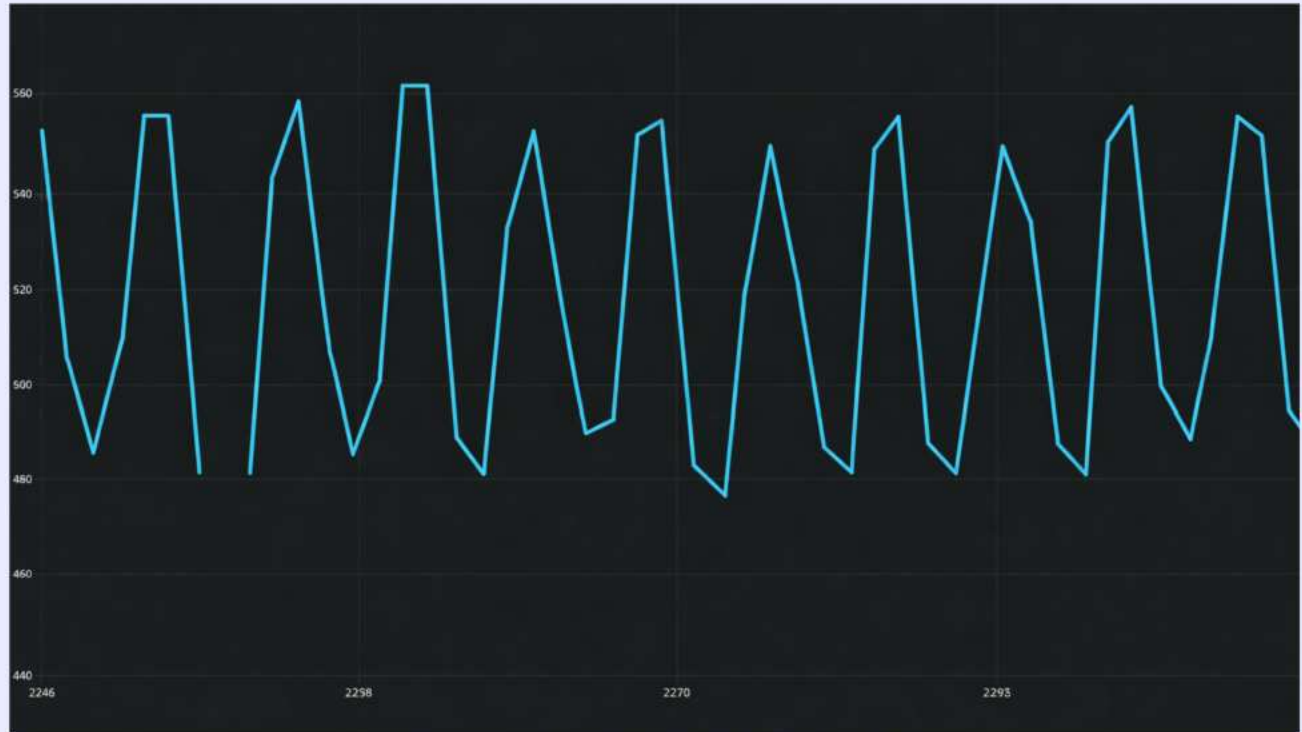


Photo created by the researcher by Arduino

Conclusion/Future

The project has been able to detect high heart rate and strong repetitive muscle contractions and classifying it as a potential seizure. The implication with this since it may be interfered with excessive workouts, I tried to outrun this by detecting for very specific repeating muscle contractions but it may be triggered during extensive workouts. To mitigate this in the future, I want to add the accelerometer to detect breathing patterns. Finally, integrating the hardware sensors: accelerometer to the software application. Further addressing this in the trends view having 3 separate lines in the graphs and displaying where they intersect to see what combinations of heart rate, muscle contractions, and breathing patterns result in a seizure.

References

All photos created by the researcher

Pedersen, M. (2023, September 21). EMG Sensors: Everything You Need to Know. IMOTIONS. <https://imotions.com/blog/learning/research-fundamentals/emg-sensors-everything-you-need-to-know/?srsltid=AfmB0orTTxu0Za5kTPNI5Whss7RmFoXq60pnz2StBqDG1wqtpqiRgWbv>

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